Lab 1: Introduction to Arduino in a simulator

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1 Student Outcomes

By the end of this lab, students will be able to:

1. Build and simulate simple electrical circuits using TinkerCAD, including the use of sensors, speakers, voltmeters, and Arduino boards.
2. Write and modify basic Arduino programs involving variables, loops, functions, conditionals, serial communication, and random number generation.
3. Develop problem-solving and independent learning skills by using online resources and AI tools to debug code, understand electronics concepts, and implement creative Arduino-based projects.

2 Equipment list

1. Tinkercad.com
2. Use any AI tool such as ChatGPT, Gemini, etc. to help you with any problem.

3 Reading/Getting Started

1. **Create an account**. If you have a TinkerCAD account, skip to the next step. If you don't have a TinkerCAD account, then go to https://www.tinkercad.com/. On the top right, click signup. Create an account, using your personal email or UIC email.
2. **Joining the class:** Go to https://www.tinkercad.com/ and log into your account. On the left toolbar, click "Classes". Then click "Join a Class". Use the code SEUXW58JQ to join the class. You will have now joined UIC-SHARP!. This is where you will submit your TinkerCAD work.
3. **Getting started with circuits**. Once you are logged in, you should be on the dashboard. If not, use this link to get to the dashboard, assuming you are logged in: https://www.tinkercad.com/dashboard. On the right side, you should see "+ New". Click

on this icon and choose “Circuit”. Now follow the instructions in this video on how to create a simple circuit using a bulb and a piezo (speaker) https://youtu.be/bpF287c-cUA. Here is a video that shows how to use an ammeter and voltmeter https://youtu.be/Uv5ATZ2JslY.

4. **Getting started with Arduino code.** Go back to the dashboard and choose circuit (see instructions above). Now choose the Arduino UNO. If it is not visible, use the search feature. Drop the Arduino UNO on the left side. Now select “Code”. By default, this is set to block code. In the drop-down bar, choose “Text”.

   (1) Write the following Arduino code:
   ```
   void setup()
   {
   Serial.begin(9600);
   Serial.print(”Hello in setup \n”);
   }
   void loop()
   {
   }
   ```
   Start the simulation. Open the “Serial Monitor” which is right below the code. You should see “Hello in setup” printed.

   (2) Write the following Arduino code:
   ```
   void setup()
   {
   Serial.begin(9600);
   }
   void loop()
   {
   Serial.print(”Hello in loop \n”);
   }
   ```
   Start the simulation. You should see “Hello in loop” printed several times. From this you can conclude that setup() function runs once and the loop() function runs continuously.
   Now add a delay by writing *delay(2000)* after Serial.print and run the code.

   (3) Write the following Arduino code that prints a variable:
   ```
   int var = 5;
   void setup()
   {
   Serial.begin(9600);
   }
   void loop()
   {
   Serial.print(var);
   Serial.print(”\n”);
   }
   ```

5. **Connecting devices and programming the Arduino:** Choose the Piezo (speaker). Connect the positive to pin 12 (Digital PWM-) on the Uno and the negative to Ground (GND). Now copy and paste the code from this webpage and start the simulation:
https://learn.adafruit.com/adafruit-arduino-lesson-10-making-sounds/playing-a-scale

4 Exercises

4.1 Arduino user input/function:

This problem tests you on the ability to take in user inputs and write a function: Write code that will take input of two numbers from the serial window (the serial window is right below the code block on TinkerCAD) and print their sum in the serial window. Use a function called sum that outputs the sum of given numbers. **HINT:** Use *Serial.read()*. See this example
https://www.arduino.cc/reference/en/language/functions/communication/serial/read/

How to submit: On TinkerCAD please submit as part of "Lab1a Arduino user input/function". The code should produce the intended output when the grader presses "simulate".

4.2 Arduino random numbers/conditionals:

This problem tests you on the ability to use random number generator and conditionals: Write code that will generate random numbers in the range of $0 - 9$ every 1 second and print it to the serial monitor if the number is less than 5. **HINT:** Use *random* to generate random numbers. See this example.
https://www.arduino.cc/reference/en/language/functions/random-numbers/random/. Use conditionals, *if-else*, for the logic.

How to submit: On TinkerCAD, please submit as part of "Lab1b Arduino random numbers/conditionals". The code should produce the intended output when the grader presses "simulate".

4.3 Arduino musical:

This problem tests you on the ability to search and utilize code from the internet: Search the internet for a musical program, and play it.

How to submit: On TinkerCAD, please submit as part of "Lab1c Arduino musical". The code should produce the intended output when the grader presses "simulate".